1. PURPOSE

This procedure describes the safe and GMP compliant operation for charging solvents to reactors from transfer booth drum pumps in the Process Development Facility (PDF) during Active Pharmaceutical Ingredient (API) and related manufacturing operations.

2. SCOPE

This procedure applies to the operational staff and describes the charging of solvents into all standard reactors R01-R22 and filter driers in the PDF, Building 902 at Sandwich, from the following sources:

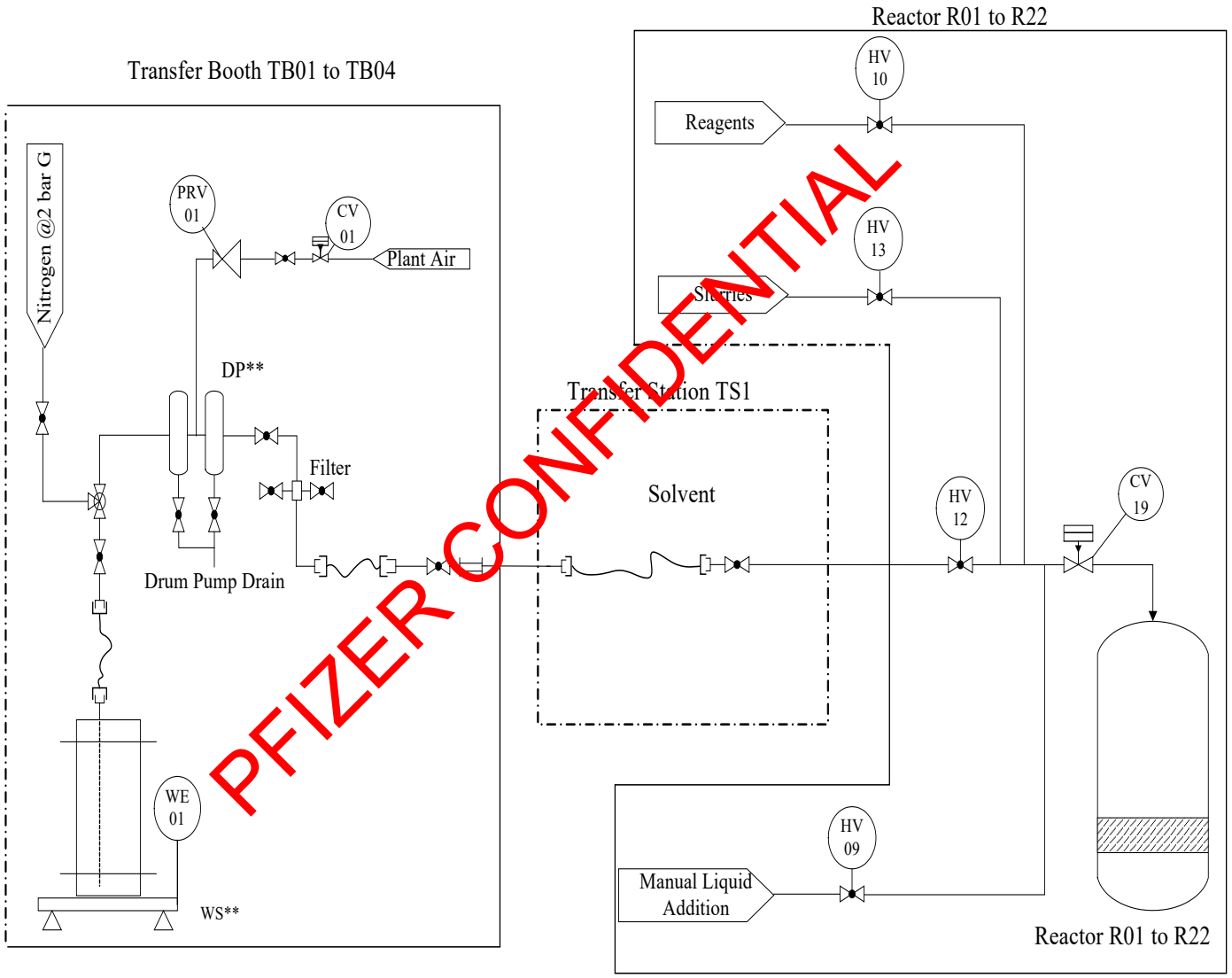
- Transfer booth TB01 drum pump DP01 to R01-R06, MFD02, MES01
- Transfer booth TB02 drum pump DP02 to R07-R12, FD03, FD04
- Transfer booth TB03 drum pump DP03 to R13-R18, FD05
- Transfer booth TB04 drum pump DP04 to R19-R22, FD02, MES04

A list of transfer destinations is supplied within each weigh scale equipment module (EM).

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3. SYSTEM DESCRIPTION

3.1 Flow Diagram



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3.2 Overview

Within each module there is an enclosed transfer booth (TB01-TB04) for dispensing liquids from drums or carboys. Each booth is equipped with a drum pump (DP01-DP04) and fixed weigh scale (0-400kg) to facilitate the transfer of solvent from the drum to the required reactor or drier. A flexible hose connection is made from the delivery line exiting the transfer booth with the required reactor solvent inlet line, at the appropriate transfer station (GP1/TS1, GP2/TS1, FG1/TS1, C1/TS1). The scale is linked via the PCS to the drum pump air supply which shuts off when the required weight has been transferred. The residual solvent in the line is then blown through to the reactor using nitrogen from the 2barg supply.

Transfers will be inhibited (CV19 closed on R01-R22) when:

- The pressure in the reactor is high
- The level in the reactor is high
- The temperature in the reactor is high
- Reactor Agitator EM is in a hold condition
- Fire Alert/Evacuate
- Temperature EM in hold condition
- Destination Manager is in a hold position
- Pressure EM is in a hold position

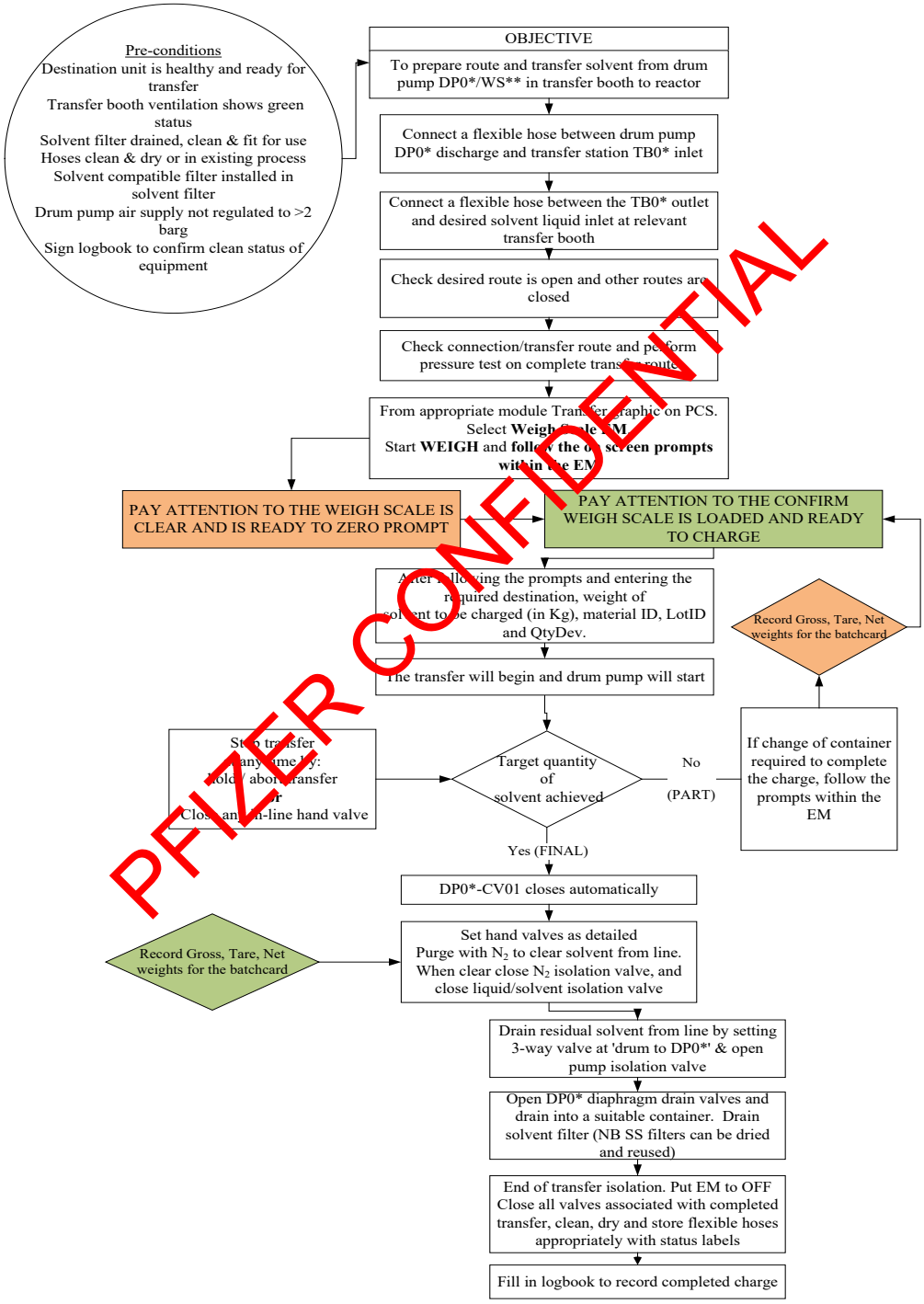
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4. SAFETY

Risk Assessment SPGRD/TB/0001752 and Manual Handling Assessment SPGRD/MH/0000235 apply to this procedure. Main areas of concern were -

- Cartridge filter changing is carried out in the transfer booth.
- Hard hats must be worn to avoid head injuries whilst working at the transfer station.
- The transfer booth drain must be capped to contain spills.
- Solvent charging to filter driers are mostly controlled via manual valves and therefore charging must always be monitored.
- The drum must be earthed when on the weigh scale and when charging.

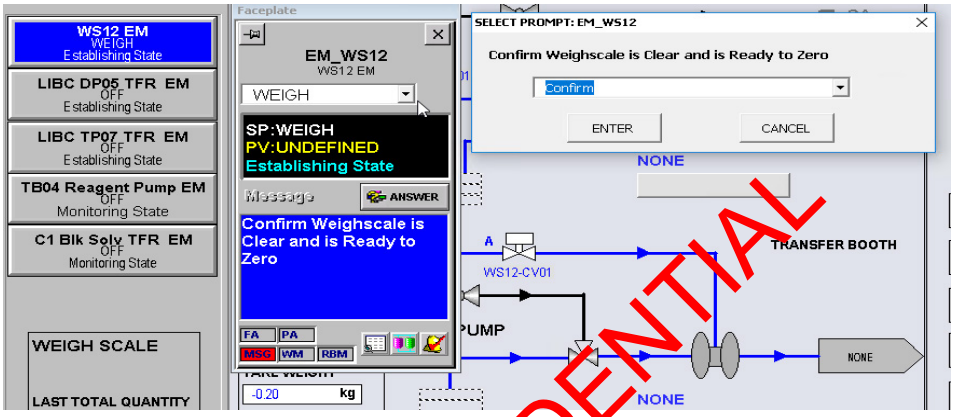
5. SOLVENT CHARGING SET-UP AND OPERATION FLOWSHEET



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5.1 Weigh scale EM Faceplate

Below is the Weigh scale EM faceplate with an example of a prompt request within the EM.



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5.2 Use of Transfer Booth Drum Pump for Non-Virgin Solvents

5.2.1 Non-virgin solvents i.e. mother liquors can also be charged via the transfer booth 1.5" transfer line to the slurry line via HV13 on the reactor.

5.2.2 No non virgin solvent charges can be made to the reactor solvent inlet line via HV12, but the SLURRY line is the preferred route.

In order to use 1.5-inch transfer line for mother liquor charges, certain preconditions need to be in place:

5.2.3 Remove stainless steel filter from inline filter housing and install a disposable filter i.e. Halar cartridge.

5.2.4 Check all equipment and lines are clean and sign logbook to confirm clean status. Label all lines, fixed and flexible with the batch number of the non-virgin solvent charge.

DO NOT USE THE ORANGE SOLVENT HOSES FOR NON-VIRGIN SOLVENT CHARGES

5.2.5 Make a logbook entry to reflect the charge.

5.2.6 Flush the line through the route to reactor or waste to remove residual liquors once transfer is complete.

5.2.7 Clean down, incorporating removal of the flexible lines from reactor slurry line and connection to waste tank.

5.2.8 Clean the route as per the cleaning sheet to obtain a Cleaning verification Test (CVT) pass before releasing pump and fixed line for the next process.

5.2.9 Label lines/pump to reflect clean status.

6. TROUBLESHOOTING

Actions on Alarms:

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Alarm	Potential Causes	Action
Reactor inlet valve R**-CV19 discrepancy/or not opening	Control valve fault / failure	Check action of control valve R**-CV19
	Reactor high level	Check level within reactor
	Reactor high pressure	Check PCS for high / high-high pressure alarm
	Reactor high-high pressure	
	Reactor high temperature	Check PCS for high / high-high temperature alarm
	Reactor high-high temperature	
	Building fire / evacuate alarm condition	Check PCS for system failures If a system failure exists refer to the relevant SOP for further action
	Pressure, Temperature or Agitator EM fault	
	Power failure	
	Instrument air failure	
Reactor manual valves alarm	More than one manual valve open	Check position of manual valves HV09, HV10, HV12 &HV13
	PCS manual valve alarm time delay too short	Check PCS time delay for manual valve alarm
Drum pump will not start	Air supply to drum pump DP0* failure	Check plant air supply to DP0* is available
	Air supply control valve DP0*-CV01 fault / failure	Check function of control valve DP0*-CV01
	Pressure reducing valve DP0*-PRV01 fault / failure	Check function of pressure reducing valve DP0*-PRV01

7. EMERGENCY ACTIONS

In the event of a fault condition with the receiving equipment or associated equipment, transfer route and associated transfer equipment the reactor inlet valve R**-CV19 will close and the plant air supply valve DP0*-CV01 to drum pump DP0* will close (pump will stop).

8. RESPONSIBILITIES

The APIM Operational Staff are responsible for the charging of solvents to reactors from transfer booth drum pump in accordance with this procedure with due consideration to the associated risks.

9. REFERENCES

SOP-EQP-00059 Use of Weigh Scales in the Process Development Facility

SPGRD/TB/0001752 Process Development Facility Generic Risk Assessment

SPGRD/MH/0000235 Liquids handling and movement in PDF

10. REVISION HISTORY

SUMMARY OF CHANGES	JUSTIFICATION OF CHANGE
1) Procedure updated as part of the Process Control System upgrade to Delta V system.	1) Procedure updated for Delta V upgrade.

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-END OF PROCEDURE-